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Homework 4: Diffractive Optical Elements and Deconvolution

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12 de noviembre de 2026

The assignment report is due on 7 October, 2026. In it, you must solve the exercises presented in the guide in report format, and attach the files you generate, such as .py, ipynb, .mat, .m, .fig, etc. The report should briefly describe the experiments performed, including the following sections: summary, description of experiments, and conclusions. It is suggested that you prepare the report in LaTeX to facilitate writing equations. Send a compressed folder with the files to the Aula submission corresponding unit: , with the subject line: **Homework4_FotografíaComputacional** and the compressed folder's name **homework4_nameSurname.zip**, where Name is your first name and Surname is your last name. Caution: any attempt at copying or fraud will invalidate the lab and result in a zero on the lab grade.

1. Learning Goals

To identify the components of a compressive color diffractive system and solving the deconvolution task. Once the homework will be completed the student will be able to simulate a diffractive color optical system and solve the deconvolution task.

2. Introduction

Computational photography involves to study optical system that exploits diffractive optical elements (DOE) to encodes the incident field.

2.1. Forward Model

The compressive difractive color measurement can be represented as:

$$\mathbf{y} = Sensor(\mathbf{h}_l * \mathbf{f}_l)_{Bayer}, \tag{1}$$

where $\mathbf{y} \in \mathbf{R}^{N\times N}$. $\mathbf{f}_l \in \mathbf{R}^{N\times N}$ denotes the lth insident field, $\mathbf{h}_l \in \mathbf{R}^{N\times N}$ denotes the point spread function (PSF), and $*$ is the convolution operation.

2.2. Deconvolution

The deconvolution process involves solving the following regularization problem

$$\min_{\tilde{\mathbf{f}}} \{ \|\mathbf{y} - Sensor(\tilde{\mathbf{f}}_l * \mathbf{h}_l)_{Bayer}\|_2^2 + \tau \|\tilde{\mathbf{f}}_l\|_2^2 \}, \tag{2}$$

τ is a regularization parameter.

2.2.1. Wiener Filter

Before to use Wiener Filter, it is obtained a demosaicked blur RGB image

$$\bar{\mathbf{f}}_l = Demosaicking(\mathbf{y}, \mathbf{B}), \tag{3}$$

once the demosaicking is performed, the Wiener filter might be applied

$$\hat{\mathbf{f}}_l = \mathcal{F}^{-1} \left\{ \frac{\mathbf{H}_l^*}{\mathbf{H}_l^2 + \tau} \bar{\mathbf{F}}_l \right\}, \tag{4}$$

where $\bar{\mathbf{F}}_l$ correspond to the measurement in the Fourier domain, $\mathbf{H}_l^*$ is the complex conjugate of the optical transfer function $\mathbf{H}_l$.

$$\tau = \frac{S_\omega}{S_f}, \tag{5}$$

where $|\mathbf{H}_l|^2 = \mathbf{H}_l^* \mathbf{H}_l$ is the magnitude squared, S_ω is the power spectrum of the noise, and S_f is the power spectrum of the signal. The complex conjugate of complex number is $C^* = R - iI$.

2.2.2. Richardson-Lucy

The Richardson-Lucy algorithm as the deconvolution algorithm, which uses an iterative approach based on back-projection of the discrepancy between the blurred color image $\bar{\mathbf{f}}_l$ and the reconstructed model. The algorithm is iterative, then the user has to select an number of iteration for all reconstructions along all color channels. The algorithm computes the reconstruction as follows

$$\hat{\mathbf{f}}_l^{k+1} = \hat{\mathbf{f}}_l^k \odot \left[\frac{\bar{\mathbf{f}}_\mathbf{l}}{\mathbf{h}_l * \hat{\mathbf{f}}_\mathbf{l}^k} * \mathbf{h}_l^\dagger \right], \tag{6}$$

where $\hat{\mathbf{f}}_l^k$ is the estimate of the original image at the kth iteration, $*$ is the convolution operation, and $\mathbf{h}_l^\dagger$ is the mirrored PSF, i.e., the spatially flipped version of the PSF used to implement the transpose of the convolution operator during deconvolution, $\odot$ is the element-wise multiplication, the division is performed element-wise.

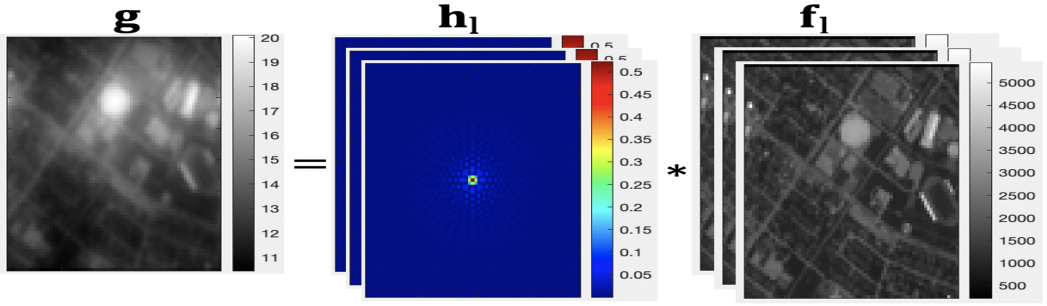


Figura 1: Sensing model

3. Exercises

1. To start, check the following Github repository; the task is to implement the MATLAB code in Python MSFA-DOE.
2. To implement a color computational photography system in Python using diffractive elements and Bayer filter. To model the diffractive optical element use the cubic phase, and also a set of Zernikes modes
3. For homework 4, use the Mc master dataset with 18 RGB images, Mc master dataset, ver Figura 2.
4. To implement from scratch both the Wiener Filter and the Richardson-Lucy algorithm and compare it with the Python implementation that uses the skimage.restoration library.
5. To evaluate spatial image quality using Peak-Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM) and spectral accuracy with Spectral Angle Mapper (SAM).
6. To plot the spatial and spectral image quality metrics for the whole Mc master dataset.



Figura 2: Mc master RGB dataset has 18 images.

The Peak Signal-to-Noise Ratio (PSNR) in decibels (dB) is given by:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right), \tag{7}$$

where MAX_I represents the maximum possible value for the image pixel (e.g., 255 for an 8-bit image). The mean squared error is given by

$$\text{MSE} = \frac{1}{m \cdot n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i,j) - K(i,j)]^2, \tag{8}$$

$$\text{SSIM}(x,y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}, \tag{9}$$

where:

- μ_x and μ_y are the local means (luminance averages) of x and y.
- σ_x^2 and σ_y^2 are the local variances (contrast) of x and y.
- σ_{xy} is the local covariance (structural similarity) of x and y.
- $C_1 = (K_1 L)^2$ y $C_2 = (K_2 L)^2$ are stabilizing constants.
- L is the dynamic range of the pixel values (typically $2^B - 1$, where B is the number of bits per pixel).

- $K_1 \ll 1$ y $K_2 \ll 1$ are small scalar contants.

$$\theta = \arccos\left(\frac{\sum_{i=1}^{n} T_i R_i}{\sqrt{\sum_{i=1}^{n} T_i^2}\sqrt{\sum_{i=1}^{n} R_i^2}}\right), \tag{10}$$

where:

- θ is the spectral angle (generally measured in radians).
- n represents the total number of spectral bands.
- T_i is the value of the test spectrum/pixel in band i.
- R_i is the value of the reference spectrum/endmember in band i.

PSNR SSIM SAM